

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

DIGITAL RADIO CONTROL MODULE (DRCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Digital Radio Control Module (DRCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: [Information and Entertainment System](#) (415-00 Information and

Entertainment System - General Information, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1A56-11	Antenna - Circuit short to ground	- Digital radio L-band antenna circuit short circuit to ground - Digital radio L-band antenna amplifier circuit short circuit to ground - Digital radio L-band antenna amplifier internal failure - Digital radio control module internal failure	- Refer to electrical circuit diagrams and check the digital radio L-band antenna circuit for short circuit to ground - Refer to electrical circuit diagrams and check the digital radio L-band antenna amplifier circuit for short circuit to ground - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio L-band antenna amplifier - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio control module
B1A56-12	Antenna - Circuit short to battery	- Digital radio L-band antenna circuit short circuit to power - Digital radio L-band antenna amplifier circuit short circuit to power - Digital radio L-band antenna amplifier internal failure	- Refer to electrical circuit diagrams and check the digital radio L-band antenna circuit for short circuit to power - Refer to electrical circuit diagrams and check the digital radio L-band antenna amplifier circuit for short circuit to power - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio L-band antenna amplifier - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio control module

		Digital radio control module internal failure	
B1A56-13	Antenna - Circuit open	- Digital radio L-band antenna circuit open circuit, high resistance - Digital radio L-band antenna amplifier circuit open circuit, high resistance - Digital radio L-band antenna amplifier internal failure - Digital radio control module internal failure	- Refer to electrical circuit diagrams and check the digital radio L-band antenna circuit for open circuit, high resistance - Refer to electrical circuit diagrams and check the digital radio L-band antenna amplifier circuit for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio L-band antenna amplifier - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio control module
B1D55-11	Antenna #2 - Circuit short to ground	- Digital radio band 3 antenna circuit short circuit to ground - Digital radio band 3 antenna amplifier circuit short circuit to ground - Digital radio band 3 antenna amplifier internal	- Refer to electrical circuit diagrams and check the digital radio band 3 antenna circuit for short circuit to ground - Refer to electrical circuit diagrams and check the digital radio band 3 antenna amplifier circuit for short circuit to ground - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio band 3 antenna amplifier - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio control

		failure Digital radio control module internal failure	module
B1D55-12	Antenna #2 - Circuit short to battery	- Digital radio band 3 antenna circuit short circuit to power - Digital radio band 3 antenna amplifier circuit short circuit to power - Digital radio band 3 antenna amplifier internal failure - Digital radio control module internal failure	- Refer to electrical circuit diagrams and check the digital radio band 3 antenna circuit for short circuit to power - Refer to electrical circuit diagrams and check the digital radio band 3 antenna amplifier circuit for short circuit to power - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio band 3 antenna amplifier - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio control module
B1D55-13	Antenna #2 - Circuit open	- Digital radio band 3 antenna circuit open circuit, high resistance - Digital radio band 3 antenna amplifier circuit open circuit, high resistance - Digital radio band 3 antenna	- Refer to electrical circuit diagrams and check the digital radio band 3 antenna circuit for open circuit, high resistance - Refer to electrical circuit diagrams and check the digital radio band 3 antenna amplifier circuit for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio band 3 antenna amplifier - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists,

		amplifier internal failure Digital radio control module internal failure	install a new digital radio control module
U2100-00	Initial Configuration Not Complete - No sub type information	Car configuration data not loaded (new module installed to vehicle and not initialized)	Using the manufacturer approved diagnostic system, re-configure the digital radio control module with the latest level software
U2101-00	Control Module Configuration Incompatible - No sub type information	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs. Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U3000-46	Control module - Calibration/parameter memory failure	Digital radio control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the digital radio control module with the latest level software
U3000-49	Control Module - Internal electronic failure	Digital radio control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio control module

U3000-98	Control Module - Component or system over temperature	▪ Digital radio control module cooling vents obstructed ▪ Digital radio control module internal failure	▪ Check that the digital radio control module cooling vents / air circulation are not obstructed. Cool the vehicle interior down by ensuring it is in the shade and have the air conditioning on cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new digital radio control module
U3003-62	Battery voltage - Signal compare failure	▪ Mismatch between the voltage at the digital radio control module and the voltage value broadcast on the CAN bus	▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111) - and compare it to battery voltage. Refer to the electrical circuit diagrams and check the digital radio control module power and ground circuits for open circuit, high resistance